



Building interpretable predictive models for pediatric hospital readmission using Tree-Lasso logistic regression

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ABSTRACT

Objectives: Quantification and early identification of unplanned readmission risk have the potential to improve the quality of care during hospitalization and after discharge. However, high dimensionality, sparsity, and class imbalance of electronic health data and the complexity of risk quantification, challenge the development of accurate predictive models. Predictive models require a certain level of interpretability in order to be applicable in real settings and create actionable insights. This paper aims to develop accurate and interpretable predictive models for readmission in a general pediatric patient population, by integrating a data-driven model (sparse logistic regression) and domain knowledge based on the international classification of diseases 9th—revision clinical modification (ICD-9-CM) hierarchy of diseases. Additionally, we propose a way to quantify the interpretability of a model and inspect the stability of alternative solutions.

Materials and methods: The analysis was conducted on >66,000 pediatric hospital discharge records from California, State Inpatient Databases, Healthcare Cost and Utilization Project between 2009 and 2011. We incorporated domain knowledge based on the ICD-9-CM hierarchy in a data driven, Tree-Lasso regularized logistic regression model, providing the framework for model interpretation. This approach was compared with traditional Lasso logistic regression resulting in models that are easier to interpret by fewer high-level diagnoses, with comparable prediction accuracy.

Results: The results revealed that the use of a Tree-Lasso model was as competitive in terms of accuracy (measured by area under the receiver operating characteristic curve—AUC) as the traditional Lasso logistic regression, but integration with the ICD-9-CM hierarchy of diseases provided more interpretable models in terms of high-level diagnoses. Additionally, interpretations of models are in accordance with existing medical understanding of pediatric readmission. Best performing models have similar performances reaching AUC values 0.783 and 0.779 for traditional Lasso and Tree-Lasso, respectfully. However, information loss of Lasso models is 0.35 bits higher compared to Tree-Lasso model.

Conclusions: We propose a method for building predictive models applicable for the detection of readmission risk based on Electronic Health records. Integration of domain knowledge (in the form of ICD-9-CM taxonomy) and a data-driven, sparse predictive algorithm (Tree-Lasso Logistic Regression) resulted in an increase of interpretability of the resulting model. The models are interpreted for the readmission prediction problem in general pediatric population in California, as well as several important subpopulations, and the interpretations of models comply with existing medical understanding of pediatric readmission. Finally, quantitative assessment of the interpretability of the models is given, that is beyond simple counts of selected low-level features.

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1. Introduction

An increased availability of pediatric health data facilitates the investigations and efforts for improved quality of care [1] and enables improvement in several areas, including the optimization of treatments, reduction of adverse events and readmission rates and earlier identification of populations in need.

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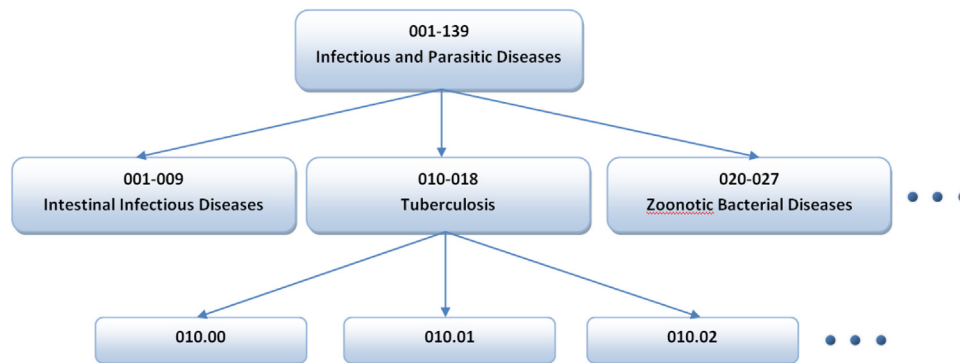


Fig. 1. Excerpt of ICD-9-CM hierarchy.

Hospital readmission (admission to a hospital within 30 days of discharge) is disruptive to both patients and healthcare providers and is often associated with higher costs and penalties. Modern care standards require effective discharge planning including the transfer of information at discharge, patient and parent education, and coordination of care after discharge. In pediatrics, the analysis of hospital readmission continues to be challenging based on the multitude of influencing factors (e.g. seasonal variations) and is considered a critical metric of the quality and cost of healthcare [2,3]. Based on a recent report [4], readmission rate within 30 days is 19.6%. Additionally, 34.0% of the pediatric patients return to the hospital within 90 days and 56.1% within one year following discharge. According to the Institute for Healthcare Improvement [4], approximately 76% of the 5 million U.S. hospital readmissions are more or less preventable [5]. Therefore, predictive modeling of the readmission risk deserves more attention to be elaborated.

Predictive algorithms can be used to identify the risk of possible readmission, and make an early warning system for the patient at the risk. Moreover, patterns in patient data as recognized by predictive algorithms could provide additional insights into the factors influencing readmission.

However, learning algorithms often fail to capture dependencies between high dimensional health related data and readmission. This could be directly related to the high-dimensionality of the data [1]. Another challenge is the high class-imbalance, meaning that majority of the patients are not readmitted within 30 days resulting in a predictive model biased in the direction of predicting a negative output (not readmitted). Another problem is that state-of-the-art predictive algorithms (that often provide highly accurate models) usually do not provide interpretable models (i.e. neural networks or support vector machines). Such models could potentially be used as early warning systems, but do not provide any actionable insights and reasons that lead to potential readmission. Finally, the notion of interpretability is subjective and in predictive modeling its analysis is most often limited to more interpretable models (i.e. Decision Trees).

This research aims to develop an accurate and interpretable predictive model for hospital readmission based on the integration of data from electronic health records (EHRs) and domain knowledge represented by the international classification of diseases 9th revision—clinical modification (ICD-9-CM) code hierarchy [6]. In this paper, we applied a Tree-Lasso logistic regression [7] in order to integrate ICD-9-CM hierarchy and logistic regression. This kind of integration of domain knowledge is interesting for research [8–10] since it improves the generalizability of predictive models and reduces the number of features, resulting in more interpretable models. Additionally, we proposed a method for quantification of interpretability of sparse predictive models.

The main contributions of this research are: (1) integration of domain knowledge (in the form of ICD-9-CM taxonomy) and

learning algorithms, to improve interpretability of models; (2) a quantitative assessment of the interpretability of predictive models, beyond simple counts of features; (3) evaluation and interpretation of predictive models for readmission prediction for State Inpatient Databases (SID) pediatric patient data in California; (4) the resulting models are interpreted for general pediatric population, as well as several important subpopulations while interpretation of models comply with existing medical knowledge.

Experiments are conducted on SID California data consisting of 67,000 pediatric patients admitted between 2009 and 2011. Each admission is described by maximum 15 diagnoses leading to over 15,000 binary features.

2. Background

Model interpretability is fairly abstract and subjective. However, a model can be defined as interpretable if the behavior of that model can be explained verbally and that the model can be used for reasoning. For medical applications, model interpretability is without any doubt a very important property, because of the underlying complexity of the phenomena that are analyzed, and the potential impact of wrong decisions. In this context, medical doctors as decision makers require a fundamental understanding of predictive models if they are implemented as clinical decision support [11].

Datasets with thousands of features are common in medicine. The interpretability of a model is related to (i) the number of features and (ii) the information provided by the features. The number of features is intuitively evident as interpretability measure. The higher the dimensionality, the more complex it becomes for human beings to analyze the relative impact of features and patterns with the potentially important in decisions. Therefore, using a reduced set of features might lead to more interpretable models. On the other hand, the contextual information provided by the features is important regardless to dimensionality. If a model is based on a limited number of features but the model is considered a black box by the human interpreter, then the model is not interpretable. This is the reason predictive modeling in the medical domain usually relies on traditional algorithms like Logistic regression (parameters of logistic regression can be interpreted as the logarithm of odds ratio) or Decision Trees (that can be interpreted as hierarchical rule set). In this research, we utilized Tree-Lasso logistic regression [7] which harnesses both elements of model interpretability. First, it forces model parameters to be zero resulting in a smaller number of features (sparse models). Second, it selects features based on group similarity (in this case to groups of diagnoses provided by ICD-9-CM codes hierarchy) facilitating the applicability of a model in a medical environment. Additionally, we proposed a method for quantification and comparison of interpretability of sparse predictive models.

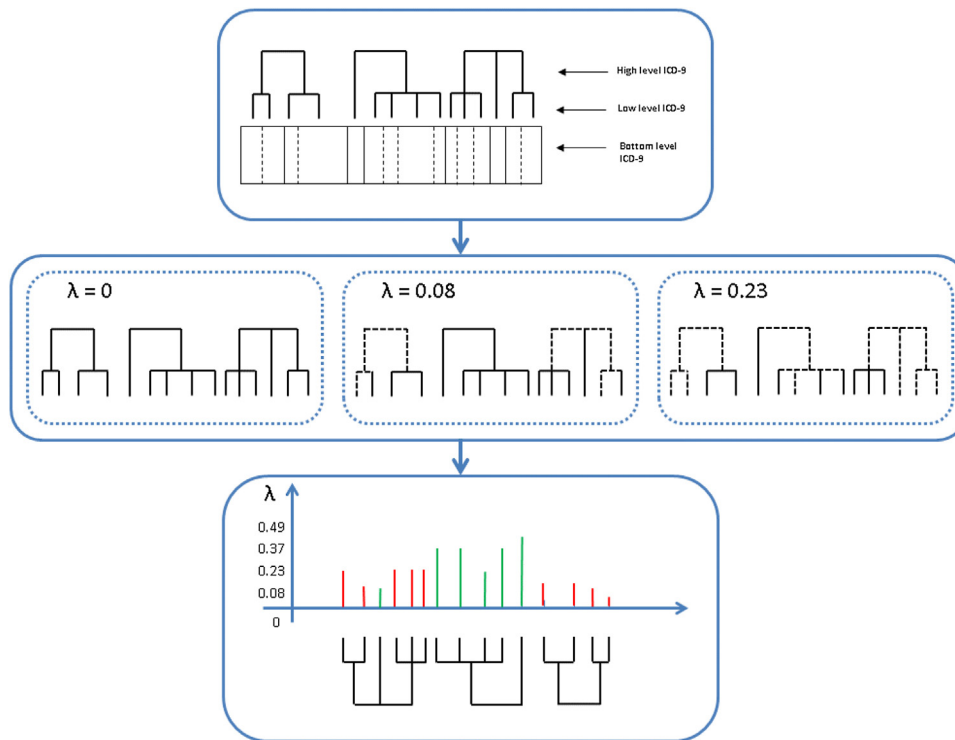


Fig. 2. Framework for development and interpretation of the Tree-Lasso logistic regression on the ICD-9-CM codes.

There are many applications based on feature selection techniques in medical tools and most of them use Lasso logistic regression. It forces most of the logistic regression weights to be equal to zero (to select fewer features) and it is often successfully implemented on highly dimensional and sparse data.

Although reducing the number of features does improve interpretability, it does so only to a degree, since selecting too little features will hurt the predictive accuracy. One solution is to apply Tree-Lasso regularization, that forces the model to select features of the same predefined (domain) groups, allowing more selected features to be interpreted by less high-level features (meaningful groups).

To the best of our knowledge, only one paper applied Tree-Lasso logistic regression for analyzing hospital readmission [10]. The goal of [10] was not to evaluate interpretability of a predictive model, but to evaluate if Tree-Lasso logistic regression was stable for feature selection. They concluded that Tree-Lasso logistic regression performed far better than other algorithms (i.e. plain logistic regression and random forests) and was more stable for feature selection compared to filter approaches (i.e. T-test, Relief, and information gain).

Despite the considerable interest [9,12–14], the effectiveness of predictive modeling focusing on 30-day hospital readmission, is weak. The area under the receiver operating characteristic curve (AUC), is currently considered to be the standard method to assess the accuracy of predictive distribution models [15]. Recently, [3] proposed an effective approach where the authors created hospital-wide predictive models. Their approach achieved AUC over 0.5 for every hospital with the best-in-class performing hospital ending up with an AUC of 0.768. By combining data from different hospitals and with the use of deep learning techniques, they managed to achieve an AUC 0.789.

3. Methods

Exploring domain knowledge based on ontologies has received significant attention recently [8,9,16]. To the best of our knowledge only one paper [10] embedded feature selection algorithms in the analysis of hospital readmission rate.

In this research, we introduce a framework for building predictive models to study and predict readmission risk. The framework also includes a method for evaluation of model interpretability. In order to exploit domain knowledge (ICD-9-CM hierarchy) for increasing interpretability, we used Tree-Lasso logistic regression [7] (also called Moreau-Yoshida logistic regression or tree-regularized logistic regression) as a predictive algorithm. The Tree-Lasso allows automatic selection of features (diagnosis) on different hierarchical levels of disease classes. This method favors models that rely on diseases from the same groups in a given hierarchy, and thus allows many selected low-level ICD codes to be interpreted in few high-level classes of diseases. This generalization of features through selection was also previously shown useful [17]. We considered the ICD-9-CM hierarchy where the structure of the features can be represented as a tree with leaf nodes as features and internal nodes as clusters of the features.

3.1. Domain hierarchy

The hierarchy employed in this paper is based on the ICD-9-CM hierarchy of diagnoses [6]. This hierarchy represents groups of diagnostic codes (called classes) ranging from a specific code to another (i.e. codes between 001 and 139 represent infectious and parasitic diseases). Additionally, each top level group is divided into one or more subgroups (i.e. infectious and parasitic diseases are subdivided into several lower level classes such as codes between 001 to 009 intestinal infectious diseases, codes between 010 and 018 present tuberculosis diagnoses etc.).

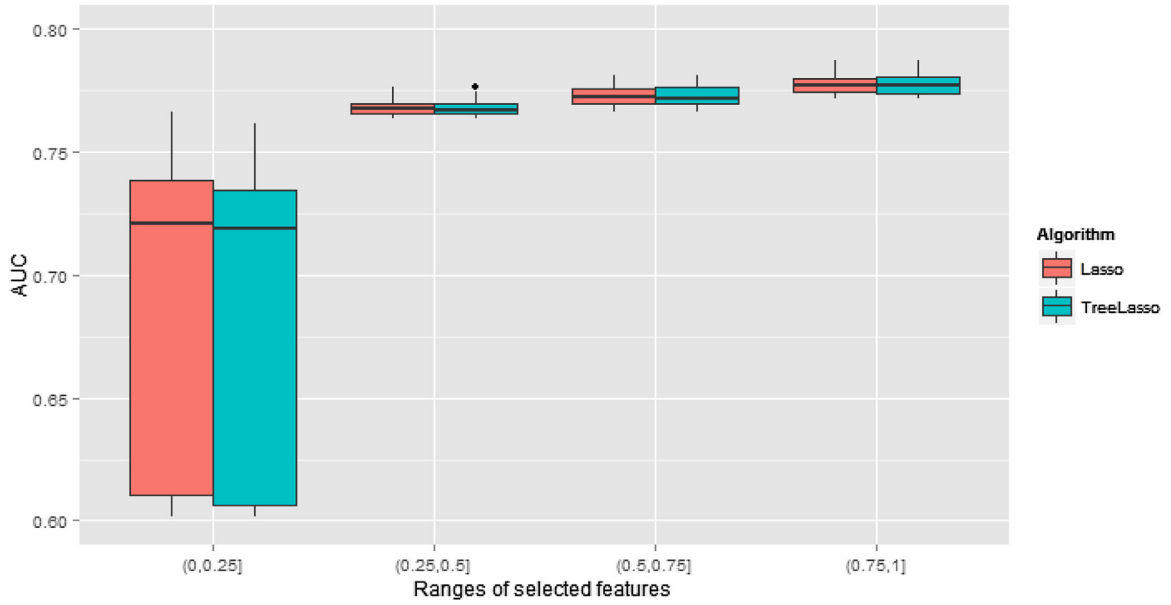


Fig. 3. Lasso and Tree-Lasso AUC performance comparison based on fraction of selected features.

Each class is connected to another by an ‘is-a’ relationship. The ICD-9-CM hierarchy enables a traverse from top level classes to bottom level classes. It is noticeable that ICD-9-CM resembles a forest (defined by a majority of trees which are not connected). A fragment of this hierarchy is presented in Fig. 1, representing a class name on the top and a corresponding ICD-9-CM code range associated with that class up to a concrete ICD-9-CM-code for diagnosis on the bottom level.

3.2. Tree-lasso logistic regression

Logistic regression is a widely used prediction model for binary outcomes. Model parameters are estimated by optimizing a cost function that penalizes deviations of model predictions from the outcomes observed in the data. Regularization is often used to limit the model complexity and prevent overfitting. Overfitting occurs when a statistical model describes random error or noise instead of the underlying relationship.

The model presented here is member of the class of regularized logistic regression models [18], which minimizes a penalized likelihood in the form:

$$L(\theta) = \sum_{i=1}^m (1 + \exp(-y_i(x^T\theta + c))) + \lambda P(\theta) \quad (1)$$

where x is a data vector, $y \in \{-1, 1\}$ is a class label, and θ is a vector with the model parameters. By imposing L1 penalty (regularization) on the parameter vector, the estimation procedure can be encouraged to prefer “sparse” parameter vectors, which is called **Lasso logistic regression** [18]. L1 penalty introduces a regularization term as: $\sum_{i=1}^n |\theta_i|$.

However, L1 regularization is oblivious to the correlations between features. Additionally, in high dimensional spaces it still selects a relatively large number of features (in our experiments over 200) diversified over different hierarchy paths leading to reduced interpretability of the models. The recently proposed **Tree-Lasso logistic regression** [7], allows leveraging our prior knowledge of the clinical connection between different ICD-9-CM codes. In particular, Tree-Lasso encourages the selection of ICD-9-CM codes that are grouped into clinically meaningful groups. This yields predictors that are more interpretable (see the results

section) and more stable, i.e., the same features are consistently selected throughout cross-validation (see Ref. [10]).

The **Tree-Lasso** uses Moreau-Yoshida regularization term:

$$P(\theta) = \sum_{i,j} w_j^i \|x_{G_i}\|_2 \quad (2)$$

where G_i is the group of features selected by the i -th node in the hierarchy, $\|\cdot\|_2$ is the L_2 norm, and w_j^i is the weight that could be assigned to each node.

The sparseness of the model was introduced with groups of features that were based on the same branch of the hierarchy and represented at a higher level in the ICD-9-CM hierarchy. As such, Tree-Lasso logistic regression can be utilized at different levels of sparsity using different regularization parameters.

In our experiments, we used publicly available implementations of both Lasso and Tree-Lasso for MATLAB [19].

3.3. Framework for hospital readmission prediction based on Tree-Lasso logistic regression

The structured regularization with a pre-defined tree structure was based on a Group-Lasso penalty, where each node in the tree is defined in one group. This regularization could facilitate the discovery in structured sparsity, which is desirable for applications with some meaningful tree structures in the features (ICD-9-CM). When estimating the model parameters, an essential step is to find appropriate values for the regularization parameter λ . A commonly used approach is to select the regularization parameter from a set of candidate values, whose values, need to be pre-specified in advance. The framework is depicted in Fig. 2. Initially, only (top level of Fig. 2) unprocessed data and ICD-9-CM codes (diagnoses) are available. In the middle layer of Fig. 2, a Tree-Lasso logistic regression was used with different values for the regularization parameter λ . Different lambda values influenced the fit uncovering the structured sparsity.

With λ equals zero, the fit is very close to the original points but with the increase of λ , the model becomes more generalized and fewer features and groups are selected. As such, with λ equals zero, no penalty is given for selecting a group of features (depicted as bottom level ICD-9-CM code in the top level of Fig. 2). This is denoted with full line edges (middle layer of Fig. 2 when λ equals 0). By increasing λ , a group penalty is introduced forcing the model to

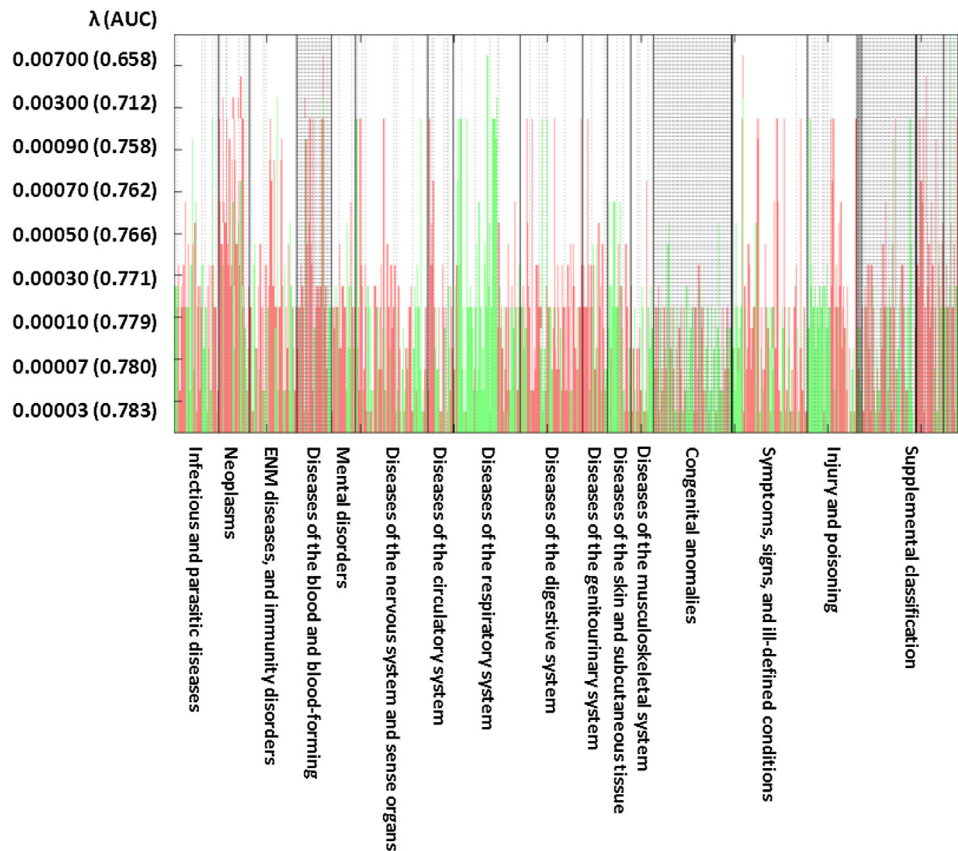


Fig. 4. Feature selection of Lasso logistic regression.

select groups of diagnoses and setting the weight of the entire group of features to zero. As a consequence, it is noticed that some groups of features are selected, i.e. the first hierarchy tree is divided into two groups (selected groups are denoted with full lines while edges of non-selected groups are presented with dashed line). Note that any given feature also represent a diagnostic group. The iteration continued to the point where λ became too large for any group to be selected (predictive model contains only the intercept).

Finally, λ (and the corresponding predictive model performance measures) in association with the features are illustrated in y and x-axis. This is depicted in the bottom layer of Fig. 2. Since a hierarchy forest exists in feature space (hierarchy consisting of multiple trees), we presented the hierarchies in the x-axis. Values are represented by every feature (x-axis), selected or not, at various values of the regularization parameter λ . By using a Tree-Lasso, an entire group of features was forced to be selected. Every feature is represented by its proper bar chart which represented the level at which this feature was considered to be important for predictive performance. The color of the bar charts indicated whether a feature positively or negatively influenced hospital readmission within 30 days, respectively in green and red color.

Subsequently, this process was repeated for data subsets containing only pediatric patients with specific diagnoses, identified as relevant for pediatric patient hospital readmission [20]. The results yielded similar patterns as the more general approach but allowed a more in-depth interpretation of models for specific diagnoses.

4. Data and experimental setup

Data was collected from hospital discharge data from the California, State Inpatient Databases (SID), Healthcare Cost and Utilization Project (HCUP) [21], Agency for Healthcare Research and

Quality. This data tracked all hospital admissions at the individual level, limiting the maximum number of diagnoses to 15 for each admission. SID provides the data format where each admission is technically limited to 15 diagnoses. Each diagnosis was represented as ICD-9-CM code. For the experimental evaluation, all pediatric patient data for a period between January 2009 and December 2011 were used.

Further, we transformed 15 features (the maximum number of diagnosis on admission) to over 15,000 binary-valued features. With over 15,000 ICD-9-CM codes (SID collects and stores data in this format), using all codes as categorical features would be highly challenging for a learning algorithm. Therefore, we transformed the diagnoses to binary features with the occurrence of diagnosis considered as a positive value, otherwise negative. Next, we excluded features with less than 50 positive values. This transformation resulted in 851 ICD-9-CM codes responsible for readmission within 30 days, as suggested in Ref. [3]. The final dataset consisted of approximately 67,000 pediatric (11,184 readmitted and 55,810 not-readmitted) patients. 30% of the data was used for testing, 70% for training. Tree-Lasso logistic regression, Lasso, and Ridge Logistic regression models were built using the SLEP package [19] in MATLAB based on the diagnostic vector on discharge. Since Ridge Logistic regression does not provide sparse models (with a low number of features) that could potentially be accurate and interpretable, we excluded this method from the analyses and focused on the comparison of Lasso and Tree-Lasso logistic regressions.

In order to assess the stability of the sparse models we repeated each experiment (for each regularization parameter) 10 times, on different random samples with repetitions, and inspected AUC distributions for different ranges of selected features.

The accuracy of the models was evaluated using area under the ROC curve (AUC) values. AUC represented the ratio of the true pos-

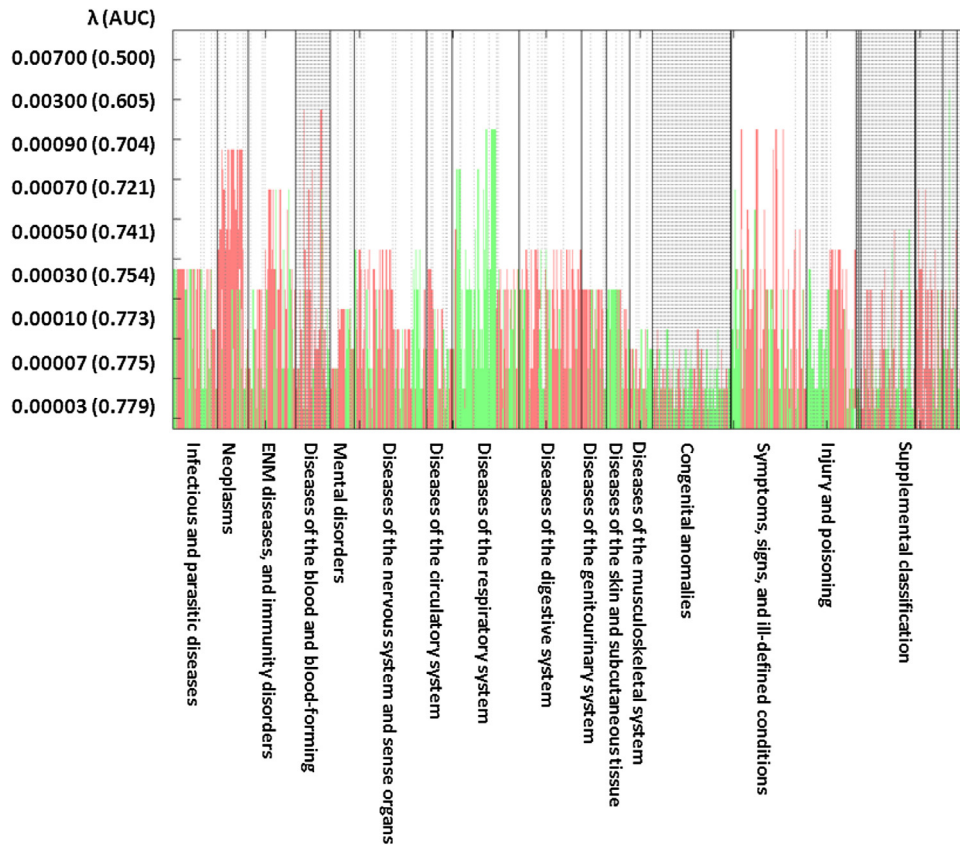


Fig. 5. Feature selection of Tree-Lasso Logistic regression.

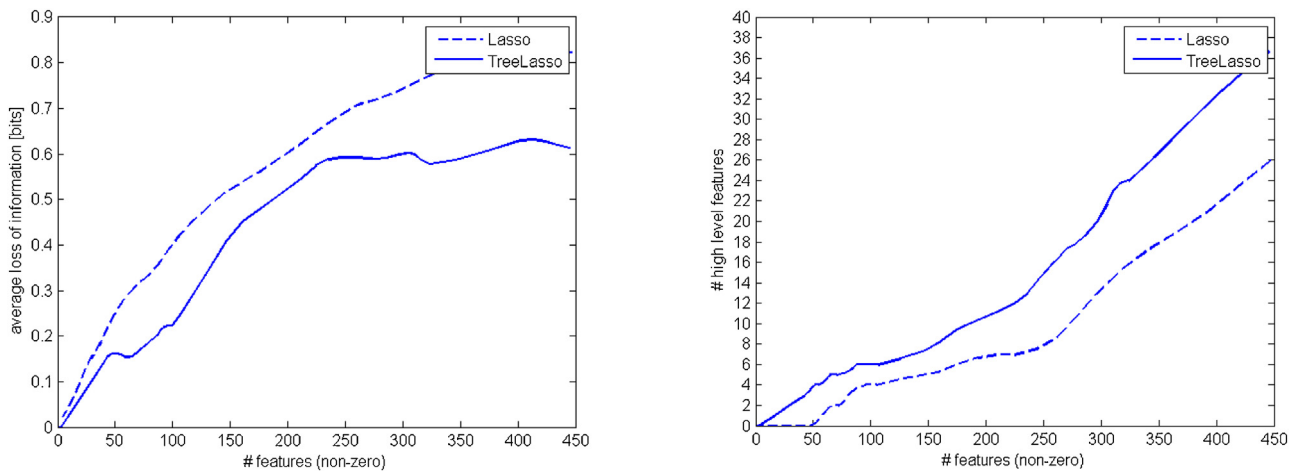


Fig. 6. (a) Information loss (left) and (b) Identified high level features (right) for different sparsity levels.

itive rate (recall) against the false positive rate (recall of negative class) at various threshold settings. By varying the threshold from 1 to 0, different values of true positive rate and false positive rate were generated. AUC have a range of [0,1], where 1 indicated that the model perfectly predicted each example at every threshold value (the true positive rate is always 1, and false positive rate is always 0). An AUC of 0.5 indicated that the predictive model was equal to chance [22].

5. Results

In the first set of experiments, we compared the performance of Lasso and Tree-Lasso models by means of AUC in relation to

the number of selected features. Lasso and Tree-Lasso algorithms were evaluated $10\times$ using different random sub-samples for training and testing. For the different λ parameter values ($n = 10, [0-1]$), a total of 100 experiments were computed, allowing a closer inspection of the stability of the models. Fig. 3 illustrates the box plots of AUC distributions (y-axis) aggregated by equidistant ranges for the percentage of features selected (x-axis). Both Lasso and Tree-Lasso models have a similar AUC performance in each range of the selected features. As we expected, by selecting only a small part of the features range [0–25%] resulted in a large variability in AUC. By selecting more features (above 25%), models become stabilized for both Lasso and Tree-Lasso. It is also interesting to notice that by adding extra features, no improvement of AUC performance is

observed (less than 0.01 when compared to features [25–50%]). Both methods were able to improve AUC scores when extra features were selected. However, using 25% of the 850 input features complicates an easy interpretation of the results. We would like to be able to explain the model in less detail, yet still use enough features to maintain a good quality of the model.

One way of interpreting the model in simpler terms is to describe the used features with the higher classes of the ICD-9-CM hierarchy. Thus, the results were sorted and grouped according to the ICD-9-CM hierarchy for both Lasso and Tree-Lasso models. Fig. 4 illustrates the selection of features by Lasso logistic regression, where the features (ICD-9-CM codes) on the x-axis are grouped so that the solid vertical lines represent boundaries of the high level in the ICD hierarchy, and dashed lines represent boundaries of the lower level (subgroups within the hierarchy) groups. The y-axis depicts the different values of the regularization parameter, so high values on y-axis correspond to model with high sparsity. The height of each line illustrates the importance of each ICD-9 code, where the highest lines are features that are introduced in the model very early and was useful even when the regularization penalty was high. Green color indicates a positive effect of features on the probability of readmission, and the red lines indicate a negative effect.

The problem with features selected by Lasso logistic regression (Fig. 4) is that it is hard to explain which features are good indicators of readmission. Even when a small percentage of features are selected, e.g. 10% of 850 features, we would need to enumerate all 85 low level selected features, and cannot interpret the model in simpler terms. If we restrict the model to very few features (by

increasing the regularization parameter), it degrades in quality too much.

Alternatively, Fig. 5 shows feature selection of Tree-Lasso logistic regression model. Visual inspection shows that low-level features tend to be selected in groups, so even with more individual features included in the model, we can describe fairly simple which higher level features the model predominantly uses. This enables the interpretation based on complete diagnostic groups, instead of specific diagnoses. Higher features are more suitable for interpretation because they are less noisy (in terms of how different physicians assign them), but also have higher support, i.e. are present more often than specific low-level ICD codes.

Based on these findings we could identify highly relevant top level groups in the diagnoses: *Neoplasms* (second from the left), and partially *Endocrine, Nutritional and Metabolic Diseases, And Immunity Disorders* group (third from the left), especially its subgroup *Other Metabolic and Immunity Disorders*. These results correlated well with clinical reality for readmissions in pediatric oncology [23] and endocrinology (diabetes) [24].

Additionally, an important diagnosis for readmissions was related to *Diseases of the Respiratory System* (8th from the left), especially its subgroup *Chronic Obstructive Pulmonary Disease and Allied Conditions* [25]. This model provided a more general prediction of readmissions in children without restriction or focus on a certain disease.

In order to validate this effect, we also ran the experiments on several subpopulations of patients and tested the interpretability of selected features which are important indicators of readmission of these subpopulations. For subpopulations, we selected the

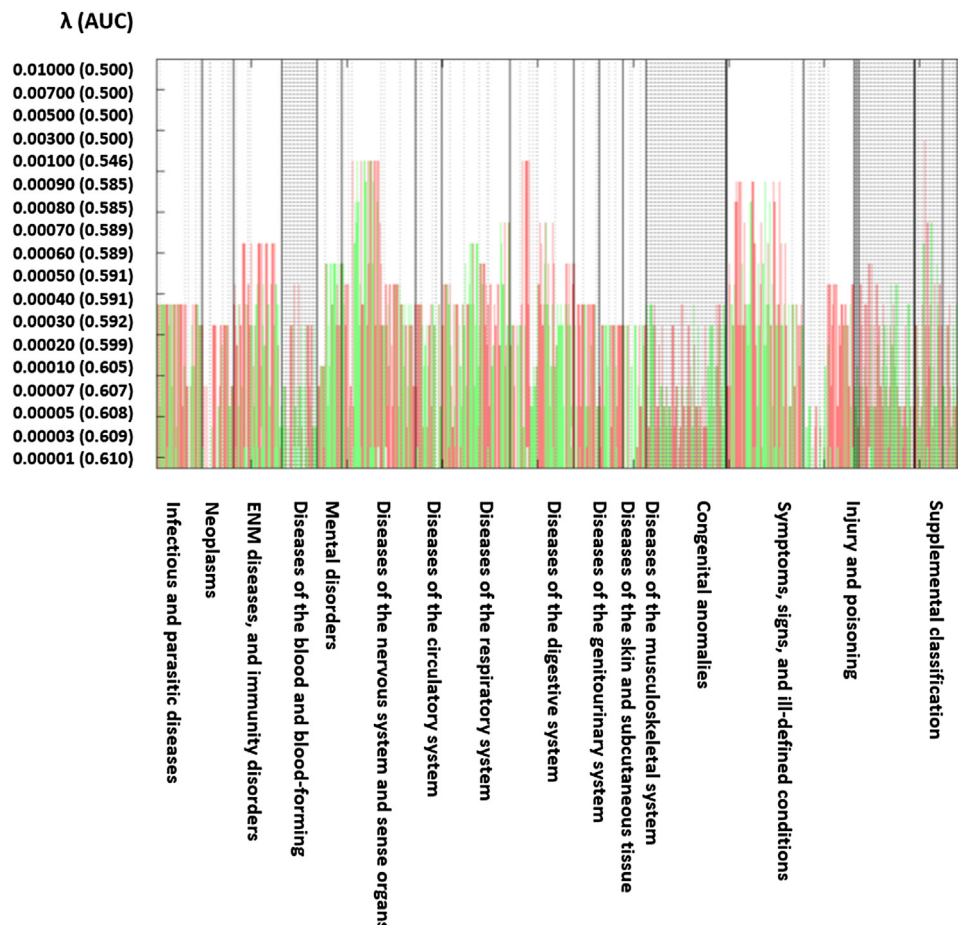


Fig. A1. Feature selection of Tree-Lasso Logistic regression for Epilepsy subpopulation.

most common pediatric diagnoses with a high prevalence of readmissions based on [20,26] paper. The results showed consistent performance, analogous to Fig. 5, and are given in the Appendix A. Next, we describe some of the interesting insights gathered from additional analyses of subpopulations of pediatric patients.

Modeling readmission in patients with Pneumonia identified these groups as highly indicative: *Poliomyelitis and Other Non-Arthropod-Borne Viral Diseases* and *Prion Diseases of Central Nervous System* (subgroup: *Other Disorders of Central Nervous System*), and also *Diseases of the Respiratory System* (*Acute Respiratory Infections*).

When modeling patient population with Epilepsy, surprisingly, diagnoses from the group *Diseases of the Digestive System* (*Diseases of Esophagus, Stomach, and Duodenum*) were highly relevant for readmission, and probably related to febrile seizures associated with temporary malnutrition [27]. Another group with high readmission was *Upper Respiratory Failure*, for which the model emphasizes patients that also had diagnoses from the subgroup *Chronic Obstructive Pulmonary Disease* and *Malignant Neoplasm of Lymphatic and Hematopoietic Tissue*.

5.1. Analysis of the interpretability

Although the interpretability is in the eye of the beholder, we can still make some objective comparisons, if not measurements, in the produced models. The first principle is that models that depend on fewer features are easier to interpret than ones with more features involved. This is the standpoint of many efforts in attribute selection and dimensionality reduction. Lasso regression is, therefore, helpful because it can reduce the feature space to only a fraction of

the original features. On the other hand, forcing the model to use too few features can lead to the poor predictive performance of the model, so feature selection can improve interpretability only to a certain level.

One way to further decrease the number of features is to extract factors from features, by mapping (transforming) the feature space to a low dimensional space, while keeping as much of the original variance of the features. While this would reduce the number of dimensions, newly produced features would be very hard to interpret, since each new feature is a combination of many original ones.

On the other hand, we can use domain hierarchies (groups) to reduce the number of features, in a way that will be interpretable for the domain experts. Instead of using the hierarchy to reduce the feature set for the model, we allow the model to use all the features but guide it using Tree-Lasso penalty to use features of the same groups. Because of this, when we interpret the selected features in terms of higher classes from the hierarchy, there will be a certain information loss because not all low-level features of the higher concept (group) would necessarily be selected.

We can measure this loss with classical information entropy [28], expressed in bits as units that depend on the relative number of selected ICD-9-CM codes in a group, i.e. the proportion of low-level ICD-9-CM codes in a group that had non-zero coefficients.

$$H = - \sum_{j=1}^q p_j \log_2(p_j) \quad (3)$$

For example, using the Tree-lasso model with 10% of original features, an interpretation that the group *Neoplasms* is important

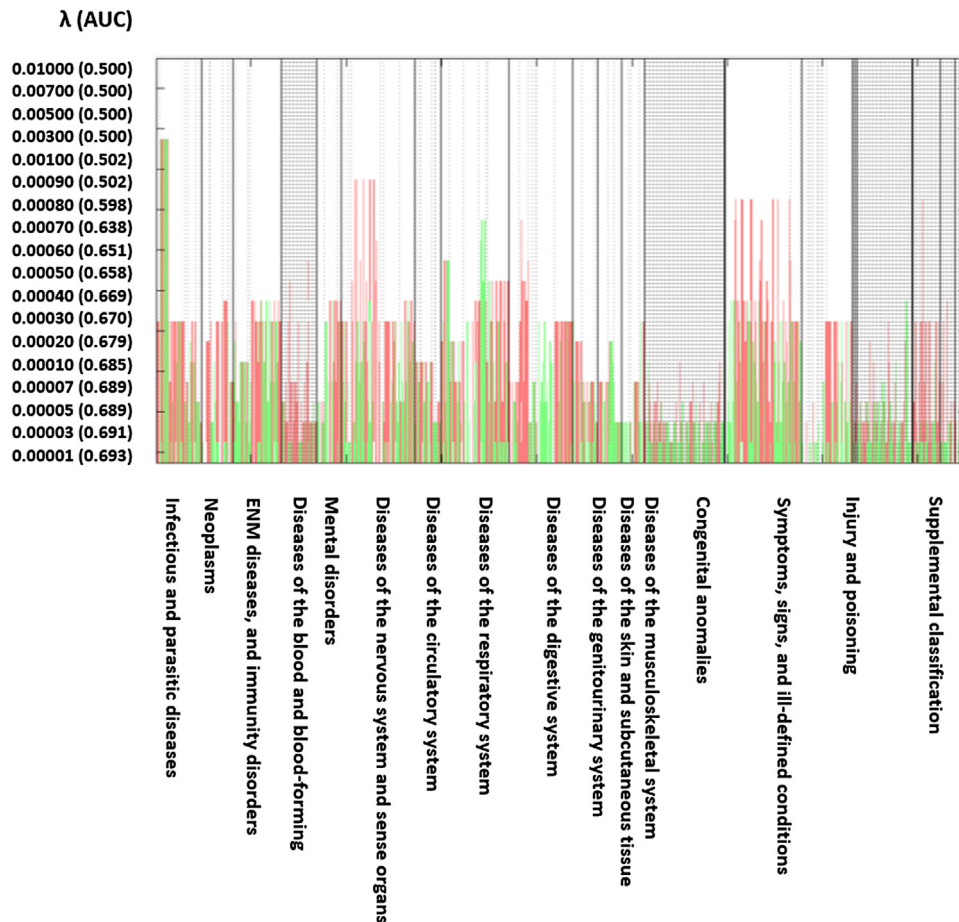


Fig. A2. Feature selection of Tree-Lasso Logistic regression for Pneumonia subpopulation.

indicator leads to information loss of 0.61 bits because the model does not select all of the ICD codes in this group (selects 28 of 33 features in the group). Similarly, the Lasso model of selection (Fig. 4) yields a loss of 0.99 bits (15 of 33 features) when the same interpretation is given. We can extend this analysis to all levels of sparsity (i.e. different amounts of selected, non-zero features), by varying the regularization parameter, which is shown in Fig. 6. Fig. 6(a) compares information loss of Lasso and Tree-Lasso (y-axis) when the same number of features is included in both models.

We notice that if we want to abstract the low-level ICD-9-CM diagnoses and interpret the model in terms of higher order ICD-9-CM groups, Tree-Lasso models always have less information loss. Also, with very sparse models, both models tend to have less information loss when we interpret them.

In Fig. 6(b), the number of high-level features (y-axis) is compared to the total number of features (x-axis) selected by both models. It can be seen that Lasso model spreads its features throughout groups, and it cannot claim that any of the groups is entirely important, where the Lasso model under 50 features has zero high-level features which it dubs relevant as a whole. On the other side, Fig. 6(b) shows that Tree-Lasso model even with 20 selected features can identify 2 high-level ICD-9-CM groups which are important. This analysis highlights the tendency of Tree-Lasso model to select ICD-9-CM codes of the same ICD-9-CM group, which aids the interpretability of important indicators for predicting readmission.

It is important to note that these differences between Lasso and Tree-Lasso and the strength of the feature groupings can be increased by modifying the w_i parameters of the regularization

penalty (Equation (2)). These parameters can be a lever to trade some more accuracy for even more interpretable models.

Finally, interpreting the models in terms of higher ICD-9-CM groups tends to produce more stable results [10,17], because low-level ICD-9-CM codes are noisy, and physicians do not agree on which exact low-level ICD-9-CM code is appropriate for each medical case. Also, expressing the model in higher concepts has more support, i.e. is covered by more instances since low-level ICD-9-CM codes can be very rare in a population.

6. Conclusion and future research

Hospital readmission in a general pediatric patient population is highly challenging machine learning task because of high dimensionality, high-class imbalance and a fact that high model interpretability is needed. We show in this paper that integration of retrospective data and domain knowledge (in the form of ICD-9-CM taxonomy) in Tree-Lasso Logistic regression can lead to improved interpretability of models for prediction of readmission. Interpretation of these models can be used for further insights of causes of future readmissions, and not only as risk indicators. Additionally, we provided a quantitative assessment of the interpretability of predictive models using information entropy and not just simple counts of features. We evaluated and interpreted predictive models for readmission prediction for SID pediatric patient data in California, both for general pediatric population and several important subpopulations. Interpretations of obtained models comply with existing medical understanding of pediatric readmission.

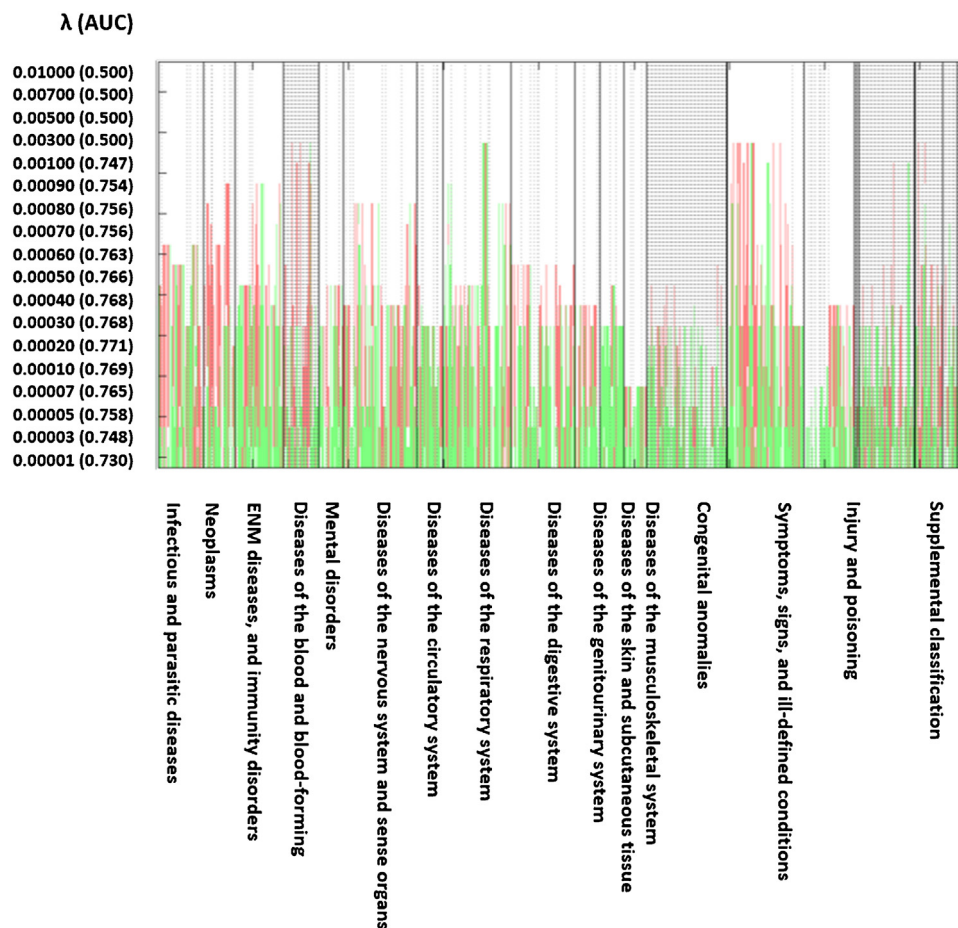


Fig. A3. Feature selection of Tree-Lasso Logistic regression for Upper respiratory tract subpopulation.

We acknowledge that enrollment in large managed care organizations may affect readmission patterns (and possibly some other confounding attributes), which can lead to different model performance for states and subpopulations not covered by this study. However, proposed framework can be adapted to other data and this methodology provides a general framework that could be used for building specific models in other states and/or subpopulations.

Additionally, other sources of domain knowledge are available. We plan to explore possibilities of enriching data driven models with different types of domain knowledge hierarchies i.e. Clinical Classification Software (CCS), Current Procedural Terminology (CPT) or diagnosis groupings i.e. Johns Hopkins ACG [29]. In the future, predictive analysis of unplanned, pediatric readmission could also be integrated into a pediatric early warning score [26].

Finally, additional patient follow-up data that is recorded after discharge should also enable a better assessment of the readmission risk and could be included in the prediction models.

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Appendix A.

See Figs. A1–A3 .

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